Lunar Life: Progress Report 2

NASA Space Apps 2026 | 14 November 2026, midday | From Aryan

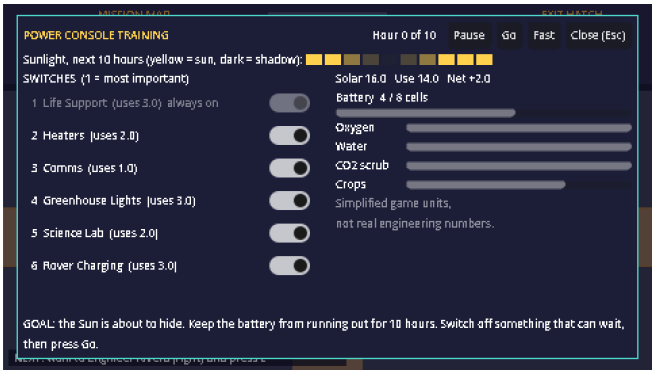
Summary

Jimmy's build brief is implemented. Lunar Life now plays from start to finish: create an astronaut, hear the briefing, pass Power Console training, walk the lunar outpost, respond to a radiation alert driven by a real archived NASA record, and read a debrief, with retry. An automated playthrough passed every check in four runs using Godot 4.7.2. A person has not yet played it end to end, and the rooms are still simple code-drawn shapes, not the look of the concept art.

What the game does now

- **Create your astronaut.** Name and seven suit colours with a live preview.
- **Briefing.** Commander Okafor gives the mission and controls before the player can move. Esc skips; talking to him replays it.
- **Training.** Engineer Rivera opens the Power Console as a panel over the room. The player must get through a scripted shadow without the battery running out. Leaving everything on fails with instant feedback and Retry; postponing a load such as rover charging passes and awards the certificate.
- **Exit hatch.** Locked until certified, then leads to the outpost.
- **Outpost.** One connected map: power room west, greenhouse east, shelter north, corridor in the middle. An objective tracker lists two routine checks. Each needs the player to walk up and press E, and each changes what is on screen.
- **Alert.** After both checks, six hours pass and the NASA event triggers the alert. Chevrons and a flashing beacon mark the route to the shelter. The storm clock runs only while the player is in control.
- **Shelter decision.** Three choices for what stays powered while the crew shelters.
- **Debrief.** Result, exposure, lowest battery, Life Support uptime, crops, science and one sentence of why, plus source notes and a retry that keeps the astronaut.



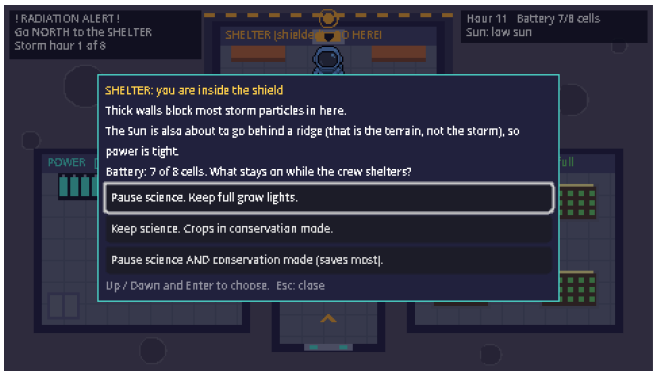


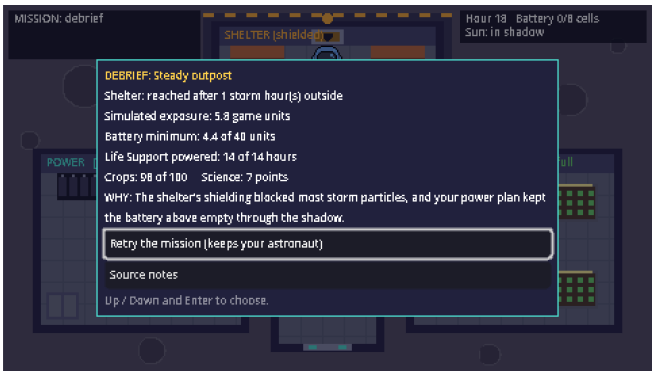
Academy room, and the Power Console training panel.





Outpost on arrival, and the alert with the route to the shelter. (The overlapping SHELTER label was fixed after these captures.)





Shelter decision, and the debrief.

NASA data

The alert uses a real record from NASA CCMC DONKI: solar energetic particle event **2024-05-10T13:35:00-SEP-001**, detected 10 May 2024 at 13:35 UTC by GOES-P: SEISS >10 MeV. It was fetched on 14 November 2026 and saved unchanged in data/nasa/donki_sep_cached.json. The game labels it as archived, states that it is a detection near Earth and not a dose at a lunar outpost, and states that the sunlight shadow comes from terrain, not the storm. Record page: https://ccmc.gsfc.nasa.gov/DONKI/view/SEP/30677/-1. There is no live fetch yet.

Test results: strategies give different outcomes

Rover policy	Shelter choice	Time outside	Exposure	Lowest battery	Outcome
Paused	Pause science and dim lamps	1 h	5.8	12.1	Steady; crops 84, science 7
Paused	Pause science, full lamps	1 h	5.8	4.4	Steady; crops 98, science 7
Charging	Pause science and dim lamps	1 h	5.8	3.9	Steady; crops 84, science 10
Charging	Pause science, full lamps	1 h	5.8	0	Life Support lost power for 1 hour
Paused	Keep science, dim lamps (arrived late)	4 h	10.7	0	Life Support lost power for 1 hour

All values are simplified game units. Each run also checked: the hatch stays locked before certification, E does nothing far from a station, walls block, the storm clock stops while a dialogue is open, repeated triggers cannot start the alert or

confirm shelter twice, and retry resets the mission while keeping the astronaut.

Status against Jimmy's acceptance checklist

Item	Status
Imports and launches without errors; customization, dialogue and training still work	Met (checked with Godot 4.7.2)
Fresh run works offline: briefing, certificate, outpost, two stations, alert, shelter, debrief	Met in the automated run
Two load strategies and a late arrival give different, explainable results	Met (table above)
Retry resets alert, tasks and resources; keeps the profile; rapid presses are safe	Met in the automated run
Prompts, warning, tracker and values legible at base resolution; not colour alone	Looks right in captures; needs a person to confirm
Room composition compared to the references	Layout and colour scheme match; detail and polish do not
Played by a person	Not yet

Known gaps

- Rooms and props are flat shapes drawn in code. Matching the concept art needs real tiles and props.
- Not built: the full Science Library, a live NASA fetch, sound, and a walk animation (walking is a bob and flip).
- Choosing panel options with the real keyboard (Up/Down and Enter) is untested; the automated run presses buttons directly.
- Nothing is committed to git yet. Only the initial commit exists.
- The team still needs to confirm which pre-event work may be submitted.
- 16 of 40 SpriteCook credits remain.

What is in this zip

Path	Contents
nasahackathon/	The Godot 4.7 project. Open project.godot and press F5.
scripts/autoload/	EventBus, GameState, ResourceSim, Controls, SceneRouter, Dialogue, MissionDirector
scripts/, scenes/	Player, mentors, academy, outpost, Power Console, panels
data/	Power loads, mentor dialogue, the NASA record and the replay settings
assets/	Four character sprites and the suit-colour shader
tools/	Automatic playthrough tool and its screenshots (tools/shots/)
docs/	CREDITS.md and both progress reports

Not included: Jimmy's handoff zip and reference images (he already has them), git history, and Godot's cache folder.